

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING			
Course Code	25MCA302	Course Type	PCC
Teaching Hours/Week (L:T:P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Mathematical Foundation for computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand the advantages and types of Artificial Intelligence		
2.	To understand machine learning, the need for machine learning, its types and data preprocessing		
3.	To understand Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning and Evaluating Regression model performance.		
4.	To understand Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms.		
5.	To learn deep learning concepts, Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks.		
UNIT - I			
Artificial Intelligence:			06 Hours
What is AI; Why Artificial Intelligence; Goals of AI; Advantages and disadvantages of AI; Applications of AI, Types of AI, Strategies for State Space Search (Representation of State Space as Graphs), Introduction to Heuristics			
UNIT - II			
Machine learning:			08 Hours
What is ML; Need for ML; Challenges and applications of ML; classification: Supervised, Unsupervised, Reinforcement learning, AI vs Machine learning, Data Preprocessing.			
UNIT - III			
Regression:			10 Hours
Simple, Linear Regression, Multiple Regression, Polynomial Regression, Logistic Regression, Decision Trees, Support Vector Machines, Naive Bayesian Learning, Evaluating Regression model performance.			
Unit - IV			
Clustering:			08 Hours
K-Means Clustering, Hierarchical Clustering, K-NN, Principal Component Analysis, Reinforcement Learning			
Unit - V			
Deep Learning and Artificial Neural Networks:			08 Hours
Introduction to deep learning, The Neuron, How do Neural network learn and work? Gradient descent, Back Propagation, ANN in python, Convolutional Neural Networks.			
Course Outcomes: At the end of the course student will be able to			
1.	Outline the fundamental concepts of Artificial Intelligence and its types.		
2.	Analyze the need for machine learning and data preprocessing		
3.	Apply Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning for data analysis and prediction.		

4.	Apply machine learning techniques like Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms for data analysis and classification.
5.	Apply deep learning concepts like Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks for classification.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA302.1	3	3	3	3	3	-	3	-	3	3
25MCA302.2	3	3	3	3	3	-	3	-	3	3
25MCA302.3	3	3	3	3	3	-	3	-	3	3
25MCA302.4	3	3	3	3	3	-	3	-	3	3
25MCA302.5	3	3	3	3	3	-	3	-	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell, "Machine Learning", 1st Edition, McGraw Hill Education, 2017.
3.	Subramanya Bhat, Machine Learning – A Simplified Approach, 1 st Edition, Narosa Publishing House Pvt. Ltd., 2025
4.	Elaine Rich, Kevin K and S B Nair, "Artificial Intelligence", 3 rd Edition, McGraw Hill Education, 2017

REFERENCE BOOKS:

1.	Saroj Kaushik, Artificial Intelligence, Cengage learning, 2014
2.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
3.	Data Science: Concepts and Practice - By Vijay Kotu, Bala Deshpande 2nd edition, Publisher -Morgan Kaufmann, 2018
4.	Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2 nd Edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016
5.	Data Science for Dummies, Lillian Pierson, John Wiley & Sons, 21-Feb-2017